

NAME

sched_get_priority_max, sched_get_priority_min – get static priority range

LIBRARY

Standard C library (*libc*, *-lc*)

SYNOPSIS

```
#include <sched.h>
```

```
int sched_get_priority_max(int policy);
```

```
int sched_get_priority_min(int policy);
```

DESCRIPTION

sched_get_priority_max() returns the maximum priority value that can be used with the scheduling algorithm identified by *policy*. **sched_get_priority_min()** returns the minimum priority value that can be used with the scheduling algorithm identified by *policy*. Supported *policy* values are **SCHED_FIFO**, **SCHED_RR**, **SCHED_OTHER**, **SCHED_BATCH**, **SCHED_IDLE**, and **SCHED_DEADLINE**. Further details about these policies can be found in **sched(7)**.

Processes with numerically higher priority values are scheduled before processes with numerically lower priority values. Thus, the value returned by **sched_get_priority_max()** will be greater than the value returned by **sched_get_priority_min()**.

Linux allows the static priority range 1 to 99 for the **SCHED_FIFO** and **SCHED_RR** policies, and the priority 0 for the remaining policies. Scheduling priority ranges for the various policies are not alterable.

The range of scheduling priorities may vary on other POSIX systems, thus it is a good idea for portable applications to use a virtual priority range and map it to the interval given by **sched_get_priority_max()** and **sched_get_priority_min()**. POSIX.1 requires a spread of at least 32 between the maximum and the minimum values for **SCHED_FIFO** and **SCHED_RR**.

POSIX systems on which **sched_get_priority_max()** and **sched_get_priority_min()** are available define **_POSIX_PRIORITY_SCHEDULING** in *<unistd.h>*.

RETURN VALUE

On success, **sched_get_priority_max()** and **sched_get_priority_min()** return the maximum/minimum priority value for the named scheduling policy. On error, *-1* is returned, and *errno* is set to indicate the error.

ERRORS**EINVAL**

The argument *policy* does not identify a defined scheduling policy.

STANDARDS

POSIX.1-2008.

HISTORY

POSIX.1-2001.

SEE ALSO

sched_getaffinity(2), **sched_getparam(2)**, **sched_getscheduler(2)**, **sched_setaffinity(2)**, **sched_setparam(2)**, **sched_setscheduler(2)**, **sched(7)**